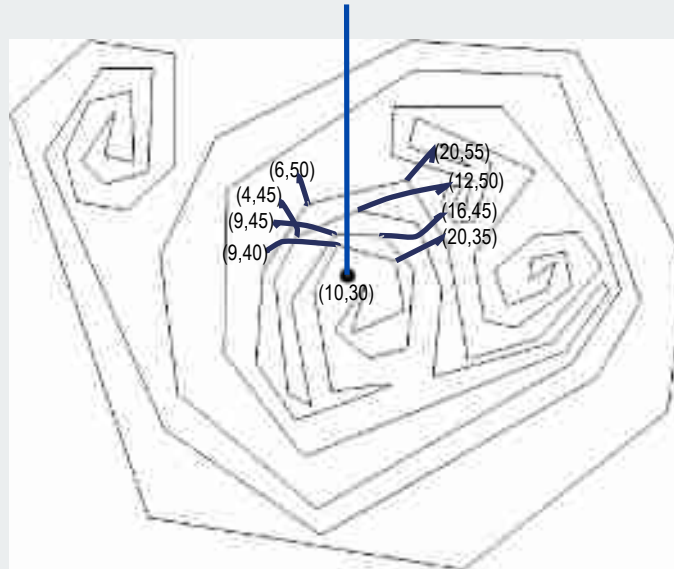


Using the coordinate of the sides given in the following figure show that these are above the point $(10, 30)$.

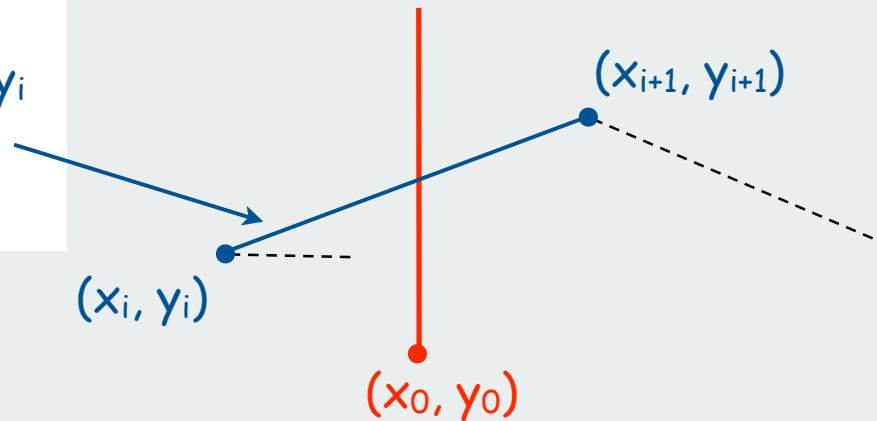


Polygon Inside, Outside: Crossing Number

Does line segment intersect ray?

$$y_0 = \frac{y_{i+1} - y_i}{x_{i+1} - x_i} (x_0 - x_i) + y_i$$

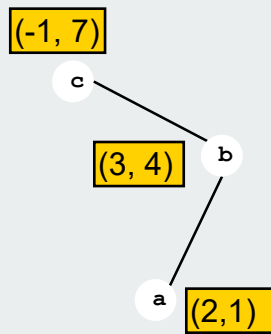
$x_i \quad x_0 \quad x_{i+1}$



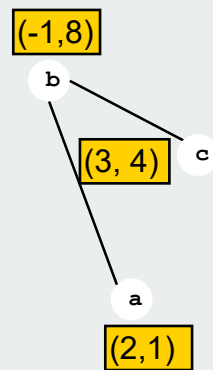
```
public boolean contains(double x0, double y0)
{
    int crossings = 0;
    for (int i = 0; i < N; i++)
    {
        double slope = (y[i+1] - y[i]) / (x[i+1] - x[i]);
        boolean cond1 = (x[i] <= x0) && (x0 < x[i+1]);
        boolean cond2 = (x[i+1] <= x0) && (x0 < x[i]);
        boolean above = (y0 < slope * (x0 - x[i]) + y[i]);
        if ((cond1 || cond2) && above) crossings++;
    }
    return ( crossings % 2 != 0 );
}
```

Implementing CCW

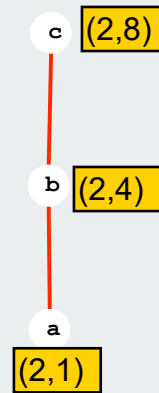
Use CCW computation technique to determine whether the points a-b-c make a CCW or not.



yes



no



???
(collinear)

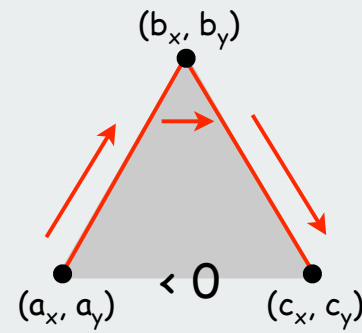
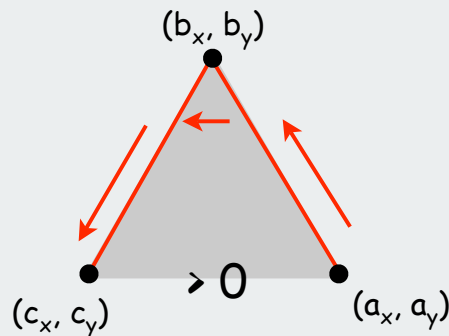
Implementing CCW

CCW. Given three point a, b, and c, is a-b-c a counterclockwise turn?

- Determinant gives twice area of triangle.

$$2 \quad \text{Area}(a, b, c) = \begin{vmatrix} a_x & a_y & 1 \\ b_x & b_y & 1 \\ c_x & c_y & 1 \end{vmatrix} = (b_x - a_x)(c_y - a_y) - (b_y - a_y)(c_x - a_x)$$

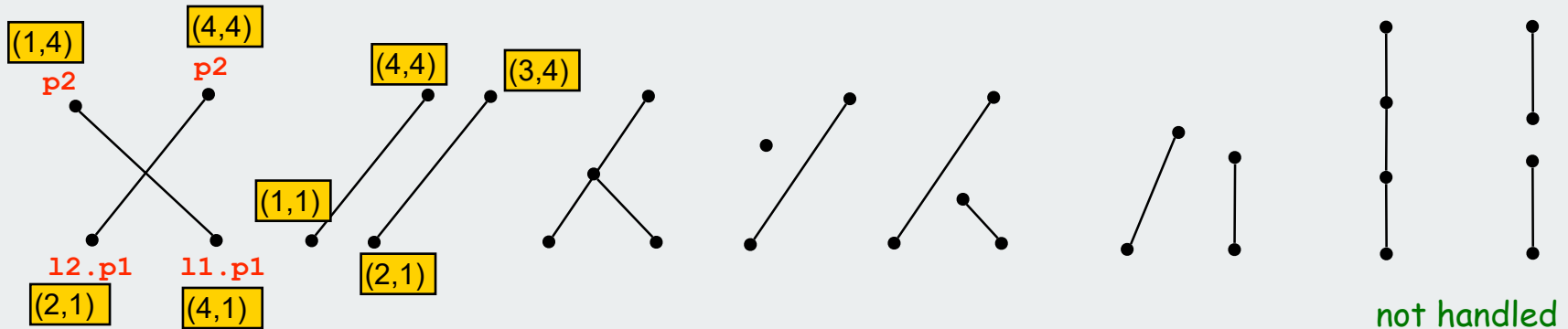
- If area > 0 then a-b-c is counterclockwise.
- If area < 0, then a-b-c is clockwise.
- If area = 0, then a-b-c are collinear.



Sample ccw client: Line intersection

Intersect: Given two line segments, do they intersect?

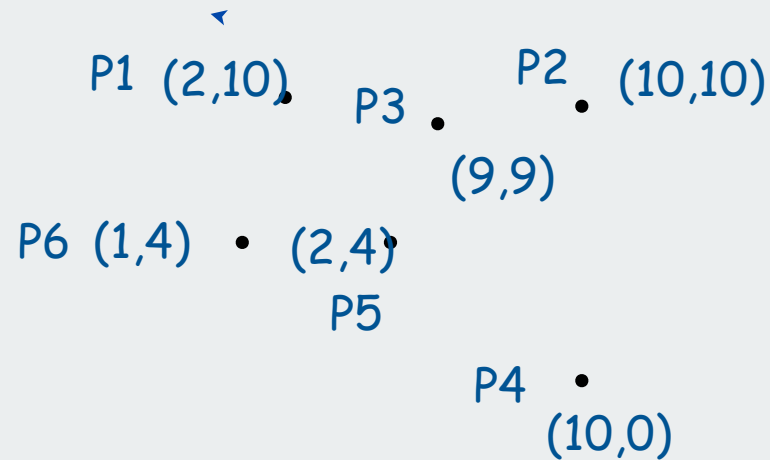
- Idea 1: find intersection point using algebra and check.
- Idea 2: check if the endpoints of one line segment are on different "sides" of the other line segment.
- 4 ccw computations.



```
public static boolean intersect(Line l1, Line l2)
{
    int test1, test2;
    test1 = Point.ccw(l1.p1, l1.p2, l2.p1)
            * Point.ccw(l1.p1, l1.p2, l2.p2);
    test2 = Point.ccw(l2.p1, l2.p2, l1.p1)
            * Point.ccw(l2.p1, l2.p2, l1.p2);
    return (test1 <= 0) && (test2 <= 0);
}
```

Use the following methods to find the convex hull for the following points

- a) Jarvis March
- b) Graham Scan Methods
- c) Brute Force Method
- d) Quick Hull Method



Closest Pair of Points

Use Divide and Conquer Approach to find the Nearest Points in the following Points. Show all the steps in determining the nearest pair of points.

$P_1(1,4)$, $P_2(4,3)$, $P_3(2,10)$, $P_4(6,7)$, $P_5(3,5)$, $P_6(8,10)$

